

Image Thresholding



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Image Thresholding

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Image Thresholding

그림하나 불러오자

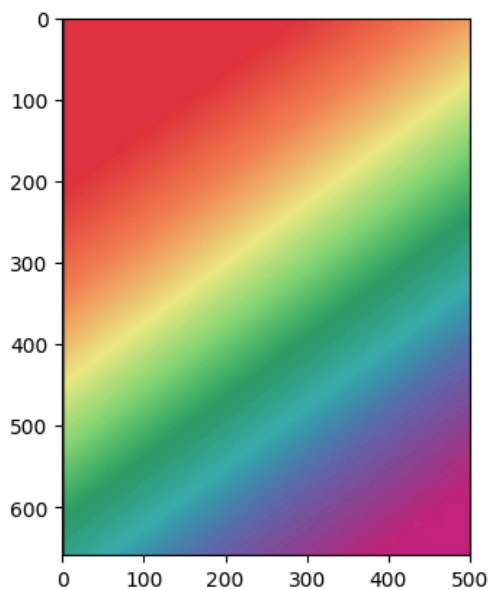
```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('./data/rainbow.jpg')
print(f'type(img) = {type(img)}, img.shape = {img.shape}')
plt.imshow(img);
```

✓ 0.3s

Python

type(img) = <class 'numpy.ndarray'>, img.shape = (659, 501, 3)

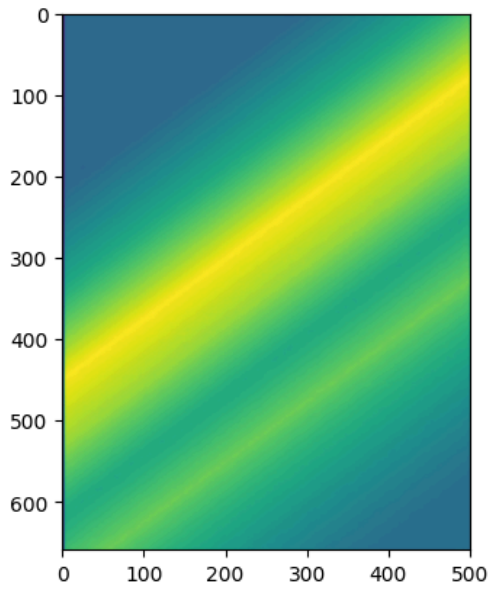


그레이 스케일로 읽으면~

```
img = cv2.imread('./data/rainbow.jpg', cv2.IMREAD_GRAYSCALE)
plt.imshow(img);
```

✓ 0.1s

Python

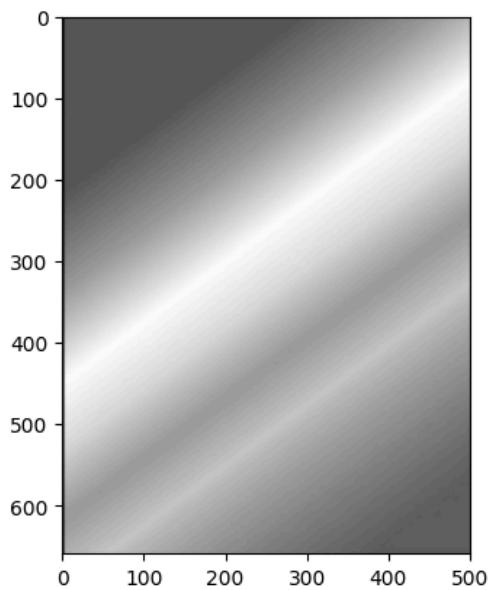


이렇게 보면 되지

```
plt.imshow(img, cmap='gray');
```

✓ 0.1s

Python



THRESH_BINARY

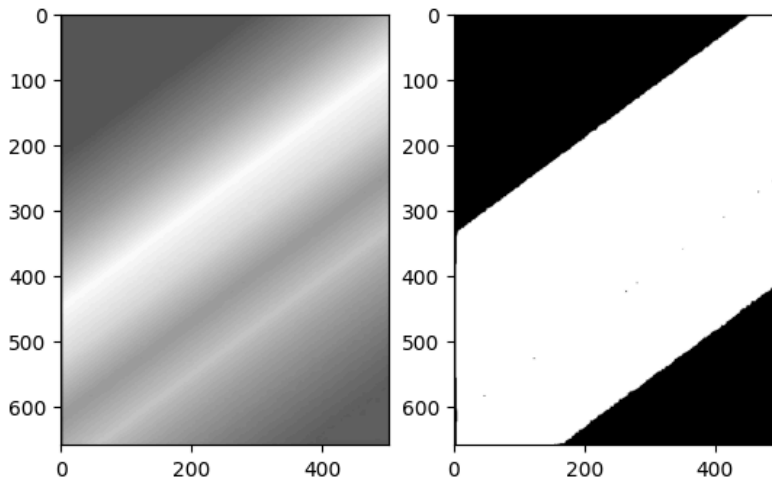
```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)
print(ret)

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.subplot(1, 2, 2)
plt.imshow(thresh1, cmap='gray')
plt.show()
```

✓ 0.1s

Python

127.0



THRESH_BINARY_INV

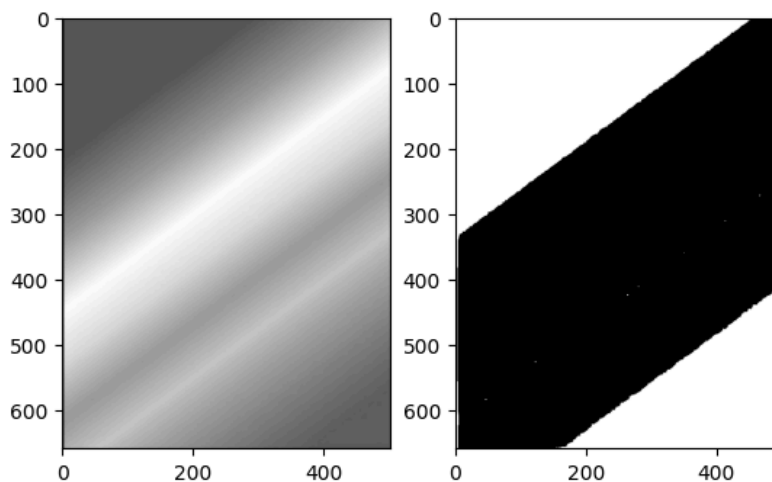
```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY_INV)
print(ret)

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.subplot(1, 2, 2)
plt.imshow(thresh1, cmap='gray')
plt.show();
```

✓ 0.1s

Python

127.0



THRESH_TRUNC

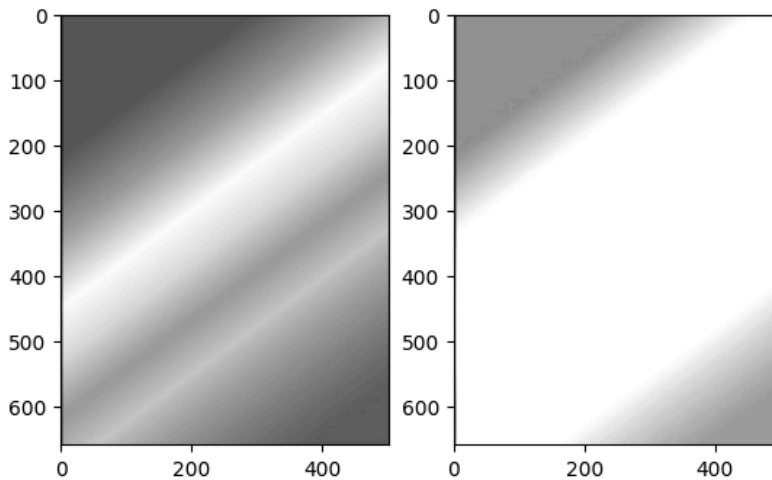
```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_TRUNC)
print(ret)

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.subplot(1, 2, 2)
plt.imshow(thresh1, cmap='gray')
plt.show();
```

✓ 0.1s

Python

127.0



THRESH_TOZERO

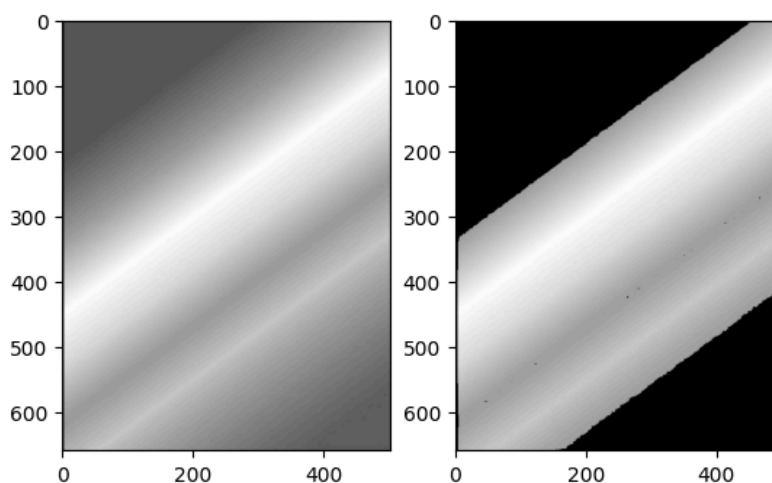
```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_TOZERO)
print(ret)

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.subplot(1, 2, 2)
plt.imshow(thresh1, cmap='gray')
plt.show();
```

✓ 0.1s

Python

127.0

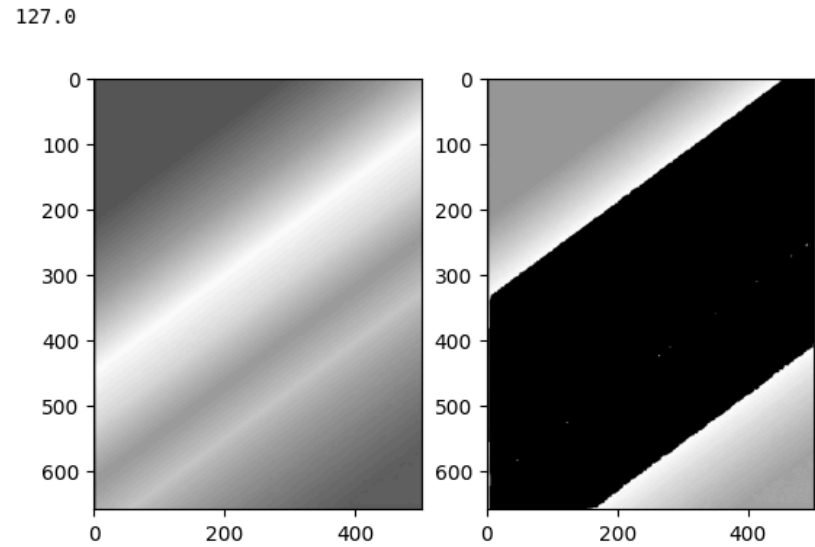


THRESH_TOZERO_INV

```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_TOZERO_INV)
print(ret)

plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.subplot(1, 2, 2)
plt.imshow(thresh1, cmap='gray')
plt.show();
```

✓ 0.1s Python



이번에는 요런 데이터 하나

```
img = cv2.imread("./data/crossword.png", cv2.IMREAD_GRAYSCALE)
print(f"type(img) = {type(img)}, img.shape = {img.shape}")
plt.imshow(img, cmap="gray")
```

✓ 0.1s Python

type(img) = <class 'numpy.ndarray'>, img.shape = (815, 515)



간단하게 그리는 함수 하나

```
def show_pic(img):  
    fig = plt.figure(figsize=(12, 12))  
    ax = fig.add_subplot(111)  
    ax.imshow(img, cmap = 'gray')
```

✓ 0.0s

Python

요런 아이

```
show_pic(img)
```

✓ 0.3s

Python

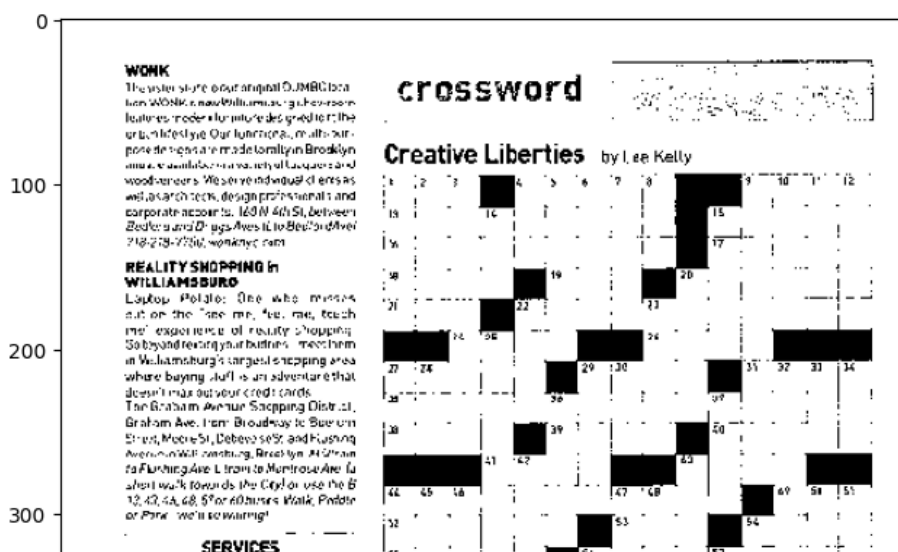


간단히 binary로

```
ret, thresh1 = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)  
show_pic(thresh1)
```

✓ 0.1s

Python



threshold 값을 바꿔볼까

```
ret, thresh1 = cv2.threshold(img, 180, 255, cv2.THRESH_BINARY)
show_pic(thresh1)
```

✓ 0.1s

Python



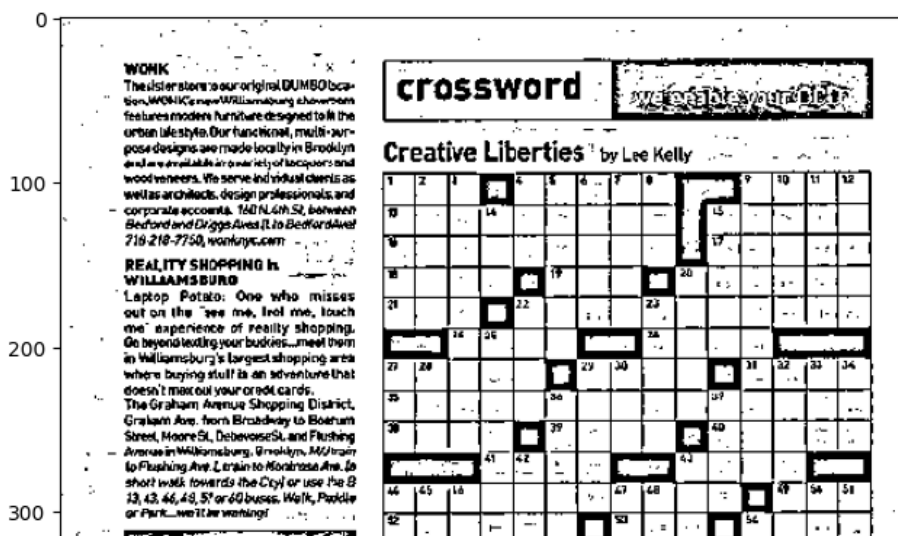
언제 또 이걸 찾나

adaptiveThreshold라는 아이가 있다

```
thresh2 = cv2.adaptiveThreshold(
    img, 255, cv2.ADAPTIVE_THRESH_MEAN_C, cv2.THRESH_BINARY, 11, 8
)
show_pic(thresh2)
```

✓ 0.2s

Python



잘 확인하기 위해 addWeighted로 겹쳐보자

```
blended = cv2.addWeighted(src1=thresh1, alpha=0.6, src2=thresh2, beta=0.4, gamma=0)
show_pic(blended)
```

✓ 0.2s

Python

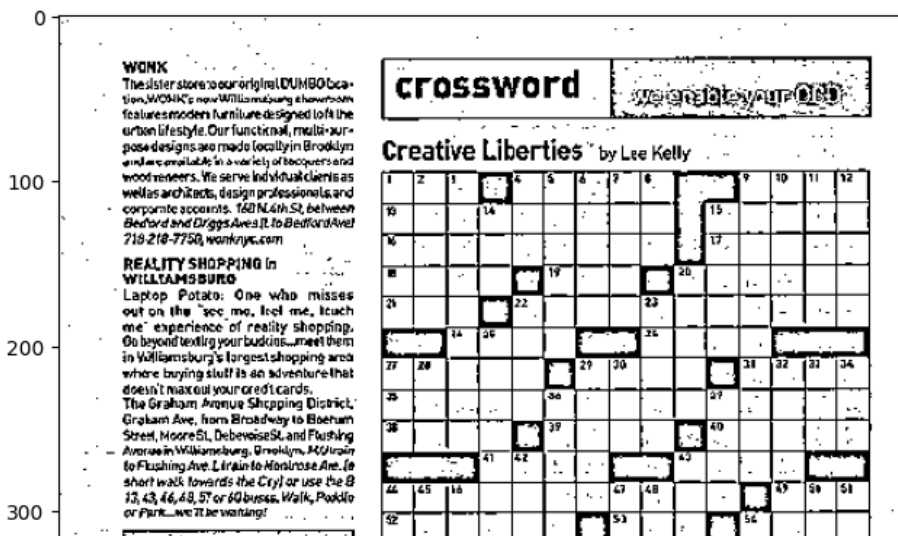


이번에는 gaussian 방법으로

```
thresh2 = cv2.adaptiveThreshold(
    | img, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY, 11, 8
)
show_pic(thresh2)
```

✓ 0.2s

Python



이번에는 이런 결과가~

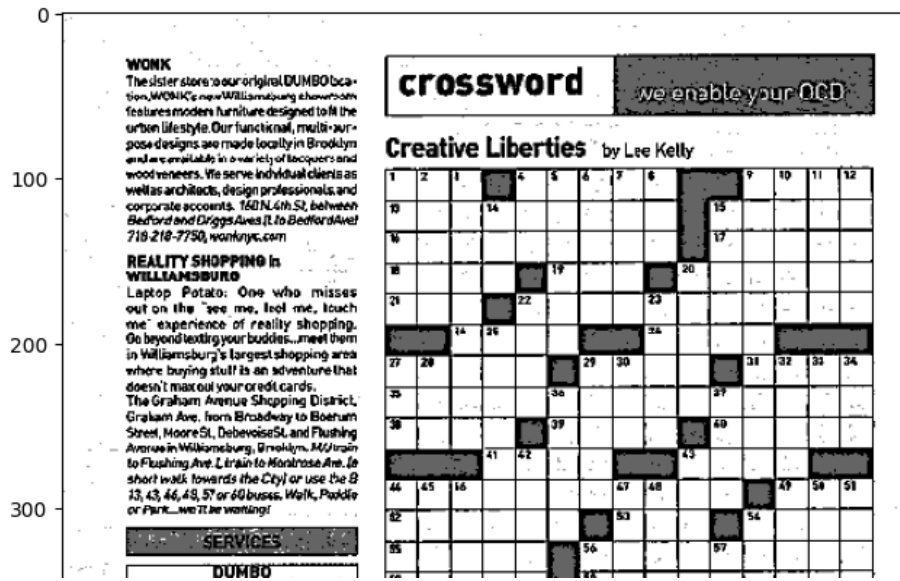
```

blended = cv2.addWeighted(src1=thresh1, alpha=0.6, src2=thresh2, beta=0.4, gamma=0)
show_pic(blended)

```

✓ 0.2s

Python





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